[Assignment-2]

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section: 28

Course Code : CSE423

Assignment - 2

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Ans to the question no . 1

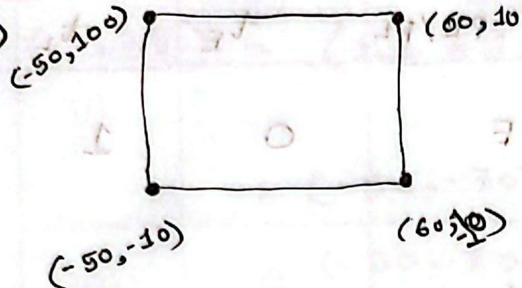
a) The AND operation in the Cohen - Sutherland Algorithm is used for trivial rejection.

If the bitwise AND of the region codes of the two endpoints of a line segment is non - zero , it indicates that both endpoints lie on the same side of the clipping window (Ex- both left , both right etc) , meaning the entire segment is outside the window and can be rejected without furthere processing.

b)
$$x(t) = x_0 + (t(x_1 - x_0)) = 1 + 0.3(7-1) = 2.8$$
$$y(t) = y_0 + (t(y_1 - y_0)) = 3 + 0.3(10-3) = 5.1$$

$$(x, y) \equiv (2.8, 5.1)$$

c)



Left boundary : $x = -50$

Right " : $x = 60$

Bottom " : $y = -10$

Top " : $y = 100$

	T	B	R	L
P1 (-50,-70) ⇒	0	1	0	0
P2 (40,100) ⇒	0	0	0	0

∴ Bitwise AND ⇒ 0 0 0 0 [So, Partially Inside]

c) A. Cyrus-Beck Algorithm :

$x_{min} = -50$

$x_{max} = 60$

$y_{min} = -10$

$y_{max} = 100$

$x_0 = -50$

$x_1 = 40$

$y_0 = -70$

$y_1 = 100$

$$D = (x_1 - x_0 , y_1 - y_0)$$
$$= (40 + 50 , 100 + 70) = (90, 170)$$

Initially, $t_E = 0$, $t_L = 1$

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Boundary	N_i	$N_i.D$	t	PE/PL	t_E	t_L
Left	(-1,0)	-90	0	PE	0	1
Right	(1,0)	90	1.22	PL	0	1
Bottom	(0,-1)	-170	0.3529	PE	0.3529	1
Top	(0,1)	170	1	PL	0.35	1

$$t_{left} = \frac{-(x_0 - x_{min})}{(x_1 - x_0)} = \frac{-(-50-(-50))}{40-(-50)} = \frac{-(-50+50)}{90} = 0$$

$$t_{right} = \frac{-(x_0 - x_{max})}{(x_1 - x_0)} = \frac{-(-50-60)}{40-(-50)} = \frac{+110}{90} = \frac{11}{9} \approx 1.22$$

$$t_{bottom} = \frac{-(y_0 - y_{min})}{(y_1 - y_0)} = \frac{-(-70-(-10))}{100-(-70)} = \frac{-(-70+10)}{170} = \frac{6\cancel{0}}{17\cancel{0}} \approx 0.3529$$

$$t_{top} = \frac{-(y_0 - y_{max})}{(y_1 - y_0)} = \frac{-(-70-100)}{100-(-70)} = \frac{170}{170} = 1$$

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$$P(0.35) = (x_0, y_0) + \frac{6}{17} \times D$$

$$= (-50,-70) + \frac{6}{17} \times (90, 170)$$

$$= (-50,-70) + \left(\frac{540}{17}, 60\right)$$

$$= \left(-\frac{310}{17}, -10\right)$$

$$P(1) = (x_0, y_0) + 1 \times D$$

$$= (-50, -70) + (90, 170)$$

$$= (40, 100)$$

$\therefore$ Line segment from $\left(-\frac{310}{17}, -10\right)$ to $(40, 100)$

c. B)

		T	B	R	L
P1 (-50,-70)	⇒	0	1	0	0
P2 (40, 100)	⇒	0	0	0	0
Bitwise AND	⇒	0	0	0	0

[Partially Inside]

P1 outcode has bottom bit.

Applying bottom intersection,

$$m = \frac{100+70}{40+50} = \frac{170}{90} = \frac{17}{9}$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} \times (x - x_1)$$

$$\Rightarrow y - 100 = \frac{17}{9} \times (x - 40)$$

$$\Rightarrow -10 - 100 = \frac{17}{9} \times (x - 40)$$

$$\Rightarrow x = -\frac{310}{17} \approx 18.24$$

		T	B	R	L
$\therefore P_1$	$\Rightarrow \left(-\frac{310}{17}, -10\right)$	0	0	0	0
P_2	$\Rightarrow (40, 100)$	0	0	0	0
Bitwise AND		0	0	0	0

$\therefore$ Line segment from $\left(-\frac{310}{17}, -10\right)$ to $(40, 100)$

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Ans to the question no 2

a) Reflection about arbitrary line vs coordinate axes

• Arbitrary line Reflection:

It requires composite transformations –

1) Translate the line to pass through the origin.

2) Rotate the line to align with a coordinate axis.

3) Reflect about that axis.

4) Apply inverse rotation and translation.

• Coordinate Axes Reflection:

1) Simple transformation with a standard matrix.

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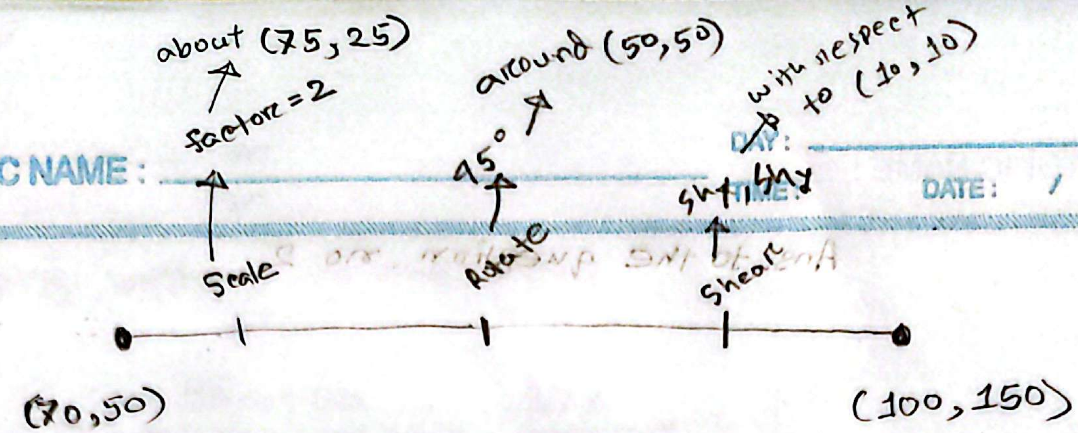
b)



$$T_1 = \begin{bmatrix} 1 & 0 & 75 \\ 0 & 1 & 25 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -75 \\ 0 & 1 & -25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & -75 \\ 0 & 2 & -25 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_2 = \begin{bmatrix} 1 & 0 & 50 \\ 0 & 1 & 50 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -50 \\ 0 & 1 & -50 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.7071 & -0.707 & 50 \\ 0.7071 & 0.7071 & -20.71 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$T_3 = \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & Sh_x & 0 \\ Sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+0 & Sh_x+0+0 & 0+0+10 \\ 0+Sh_y+0 & 0+1+0 & 0+0+10 \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & Sh_x & 10 \\ Sh_y & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+0 & 0+Sh_x+0 & \cancel{-10}-10Sh_x+\cancel{10} \\ Sh_y+0+0 & 0+1+0 & -10Sh_y-\cancel{10}+\cancel{10} \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & Sh_x & -10Sh_x \\ Sh_y & 1 & -10Sh_y \\ 0 & 0 & 1 \end{bmatrix}$$

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Let,

$$P' = T_3 \times T_2 \times T_1 \times P$$

$$= \begin{bmatrix} 1 & Sh_x & -10Sh_x \\ Sh_y & 1 & -10Sh_y \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0.7071 & -0.707 & 50 \\ 0.7071 & 0.7071 & -20.71 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\times \begin{bmatrix} 2 & 0 & -75 \\ 0 & 2 & -25 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 70 \\ 50 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & Sh_x & -10Sh_x \\ Sh_y & 1 & -10Sh_y \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 42.9365 \\ 78.284 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 42.9365 + 78.284Sh_x - 10Sh_x \\ 42.9365Sh_y + 78.284 - 10Sh_y \\ 0 + 0 + 1 \end{bmatrix}$$

$$= \begin{bmatrix} 42.9365 + 68.284Sh_x \\ 78.284 + 32.9365Sh_y \\ 1 \end{bmatrix}$$

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So,

$$\begin{bmatrix} 100 \\ 150 \\ 1 \end{bmatrix} = \begin{bmatrix} 42.9365 + 68.284\, Sh_x \\ 78.284 + 32.9365\, Sh_y \\ 1 \end{bmatrix}$$

$\therefore 42.9365 + 68.284\, Sh_x = 100$

$\therefore Sh_x = 0.84$

$78.284 + 32.9365\, Sh_y = 150$

$\therefore Sh_y = 2.18$

Ans: $Sh_x = 0.84$

$Sh_y = 2.18$

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